WEEK-1 ASSIGNMENT
Introduction to artificial intelligence & python

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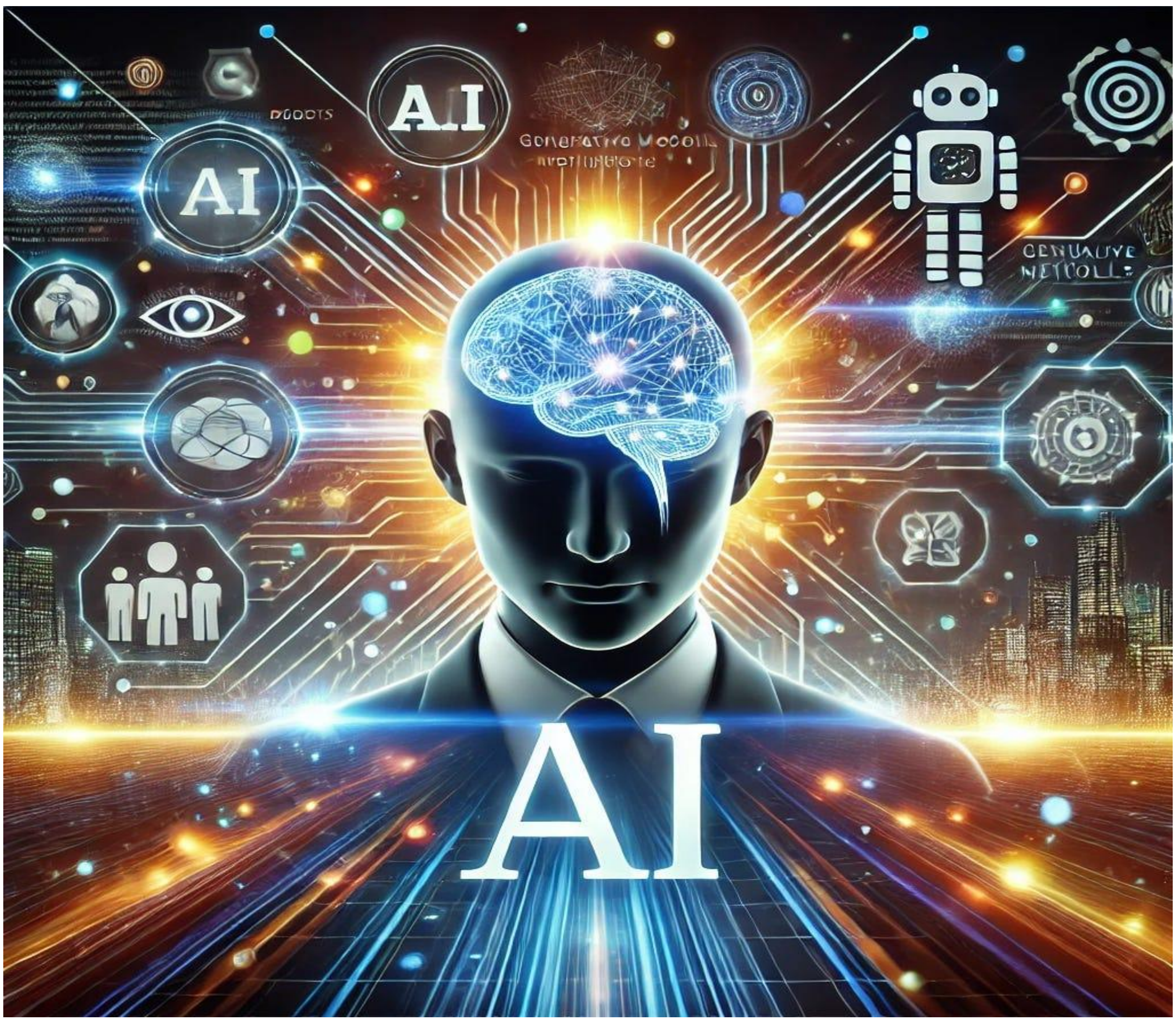
TASK-1: INTRODUCTION TO ARTIFICIAL INTELLIGENCE & PYTHON

Artificial Intelligence, commonly known as AI, is a technology that allows computers and machines to perform tasks that normally require human intelligence.
These tasks include understanding language, recognizing images, finding patterns, solving problems, and making decisions.
We already use AI in many parts of our daily life.
For example, YouTube recommends videos, Google Maps suggests routes, and smartphones can recognize faces. These systems use AI to understand information and provide useful results.
In simple words,
AI helps machines perform tasks in a smarter way by using data, rules, and computer programs.



2.TYPES OF ARTIFICIAL INTELLIGENCE

AI is generally divided into three types: **ANI, AGI, and ASI**.

2.1 Artificial Narrow Intelligence (ANI)

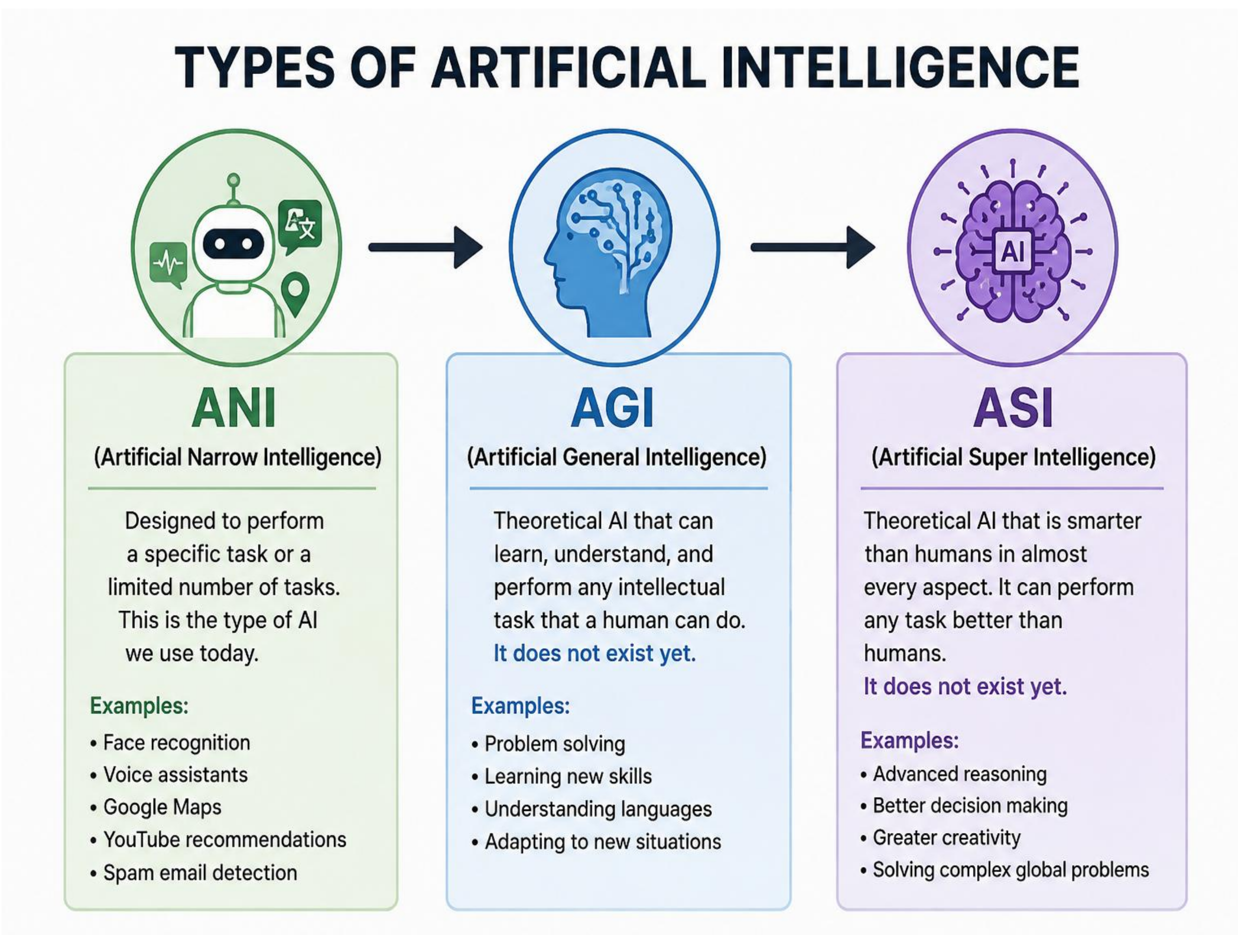
ANI is designed to perform a specific task. It cannot do everything that a human can do.
Examples: Face recognition, Google Maps, voice assistants, and YouTube recommendations.

2.2 Artificial General Intelligence (AGI)

AGI is a theoretical type of AI that would be able to learn and perform different tasks like a human. **AGI has not been achieved yet.**

2.3 Artificial Super Intelligence (ASI)

ASI is a theoretical form of AI that would be more intelligent than humans in almost every area. **ASI does not currently exist.**



3. AI vs Machine Learning vs Deep Learning
AI, Machine Learning, and Deep Learning are related concepts, but they are not the same.

Artificial Intelligence (AI) is the broad concept of making machines perform tasks that normally require human intelligence.

Machine Learning (ML) is a part of AI where computers learn patterns from data instead of being programmed for every situation.

Deep Learning (DL) is a part of Machine Learning that uses neural networks with multiple layers to learn from large amounts of data.

The relationship is:
AI → Machine Learning → Deep Learning

Technology	**Meaning**	**Example**
Artificial Intelligence	Makes machines perform intelligent tasks	Voice Assistant
Machine Learning	Learns patterns from data	Movie Recommendation
Deep Learning	Uses multi-layer neural networks	Face Recognition

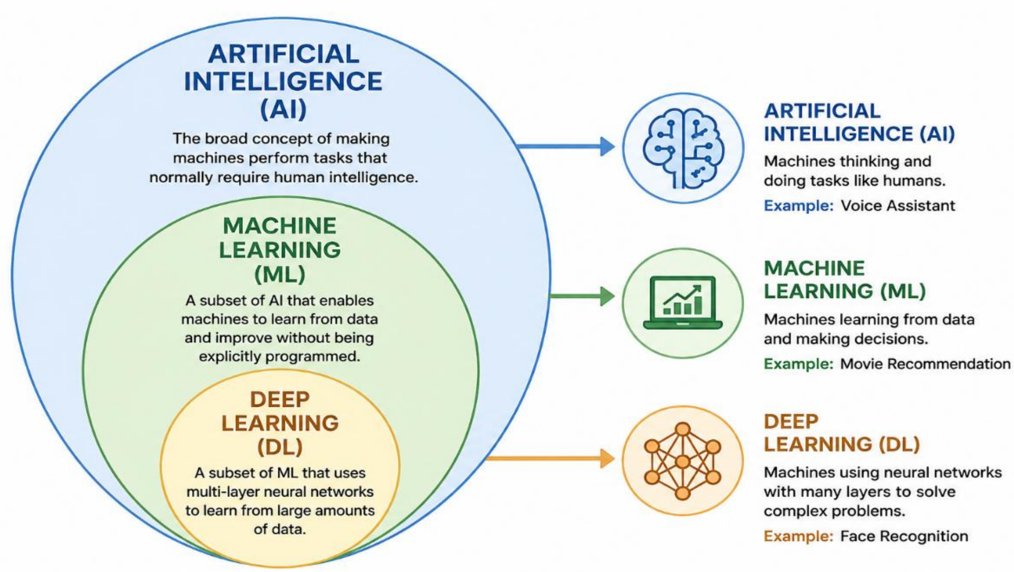


Figure 3: Relationship between AI, Machine Learning and Deep Learning

4. Real-Life Applications of AI

AI is used in many areas of our daily life. Some common applications are:

1. Healthcare: AI helps doctors analyze medical images and patient data.

2. Education: AI can provide personalized learning, explanations, and practice questions.

3. Banking: AI is used for fraud detection, transaction analysis, and customer support.

4. Entertainment: YouTube, Netflix, and other platforms use AI to recommend content.

5. Transportation: AI is used in navigation, traffic prediction, and driver-assistance systems.

6. Smartphones: AI is used for face recognition, voice assistants, camera features, and photo organization.

7. E-Commerce: Online shopping websites use AI to recommend products and improve search results.

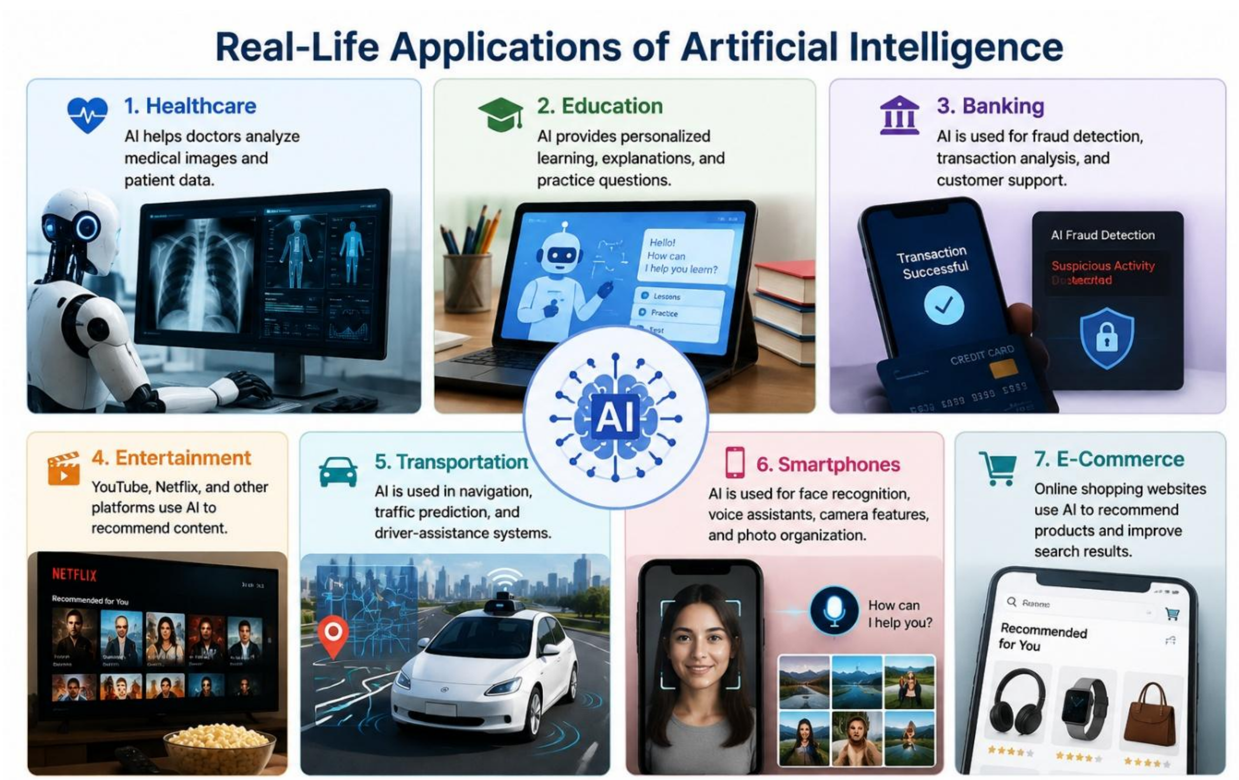


Figure 4: Real-Life Applications of Artificial Intelligence

Advantages and Limitations of AI

5. Advantages of Artificial Intelligence

1. **Automation:** AI can perform repetitive tasks automatically.
2. **Fast Processing:** AI can process large amounts of data quickly.
3. **24/7 Availability:** AI systems can work continuously.
4. **Improved Accuracy:** AI can reduce errors in many data-based tasks.
5. **Better Decisions:** AI can analyze data and provide useful information.

6. Limitations of Artificial Intelligence

1. **High Cost:** Developing and maintaining AI systems can be expensive.
2. **Data Requirement:** AI systems often need large amounts of quality data.
3. **Privacy Concerns:** AI may process personal information.
4. **Limited Human Understanding:** AI does not understand emotions like humans.
5. **Incorrect Results:** AI can sometimes give wrong results.

7. Conclusion

- Artificial Intelligence has become an important part of modern life.
- It is useful in healthcare, education, banking, entertainment, transportation, and many other fields.
- Although AI provides many benefits, it also has limitations such as cost, privacy concerns, and dependence on data.
- Therefore, AI should be developed and used responsibly.

Task-2 Setting Up AI Development Environment

Objective

The objective of this task is to set up and verify a Python-based development environment for Artificial Intelligence programming.

Tools and Libraries

Python– Primary programming language for AI development.
Visual Studio Code (VS Code) – Used for writing and running Python programs.
Jupyter Notebook – Used for interactive Python programming and data analysis.
NumPy – Used for numerical calculations and arrays.
Pandas – Used for data manipulation and analysis.

Environment Setup

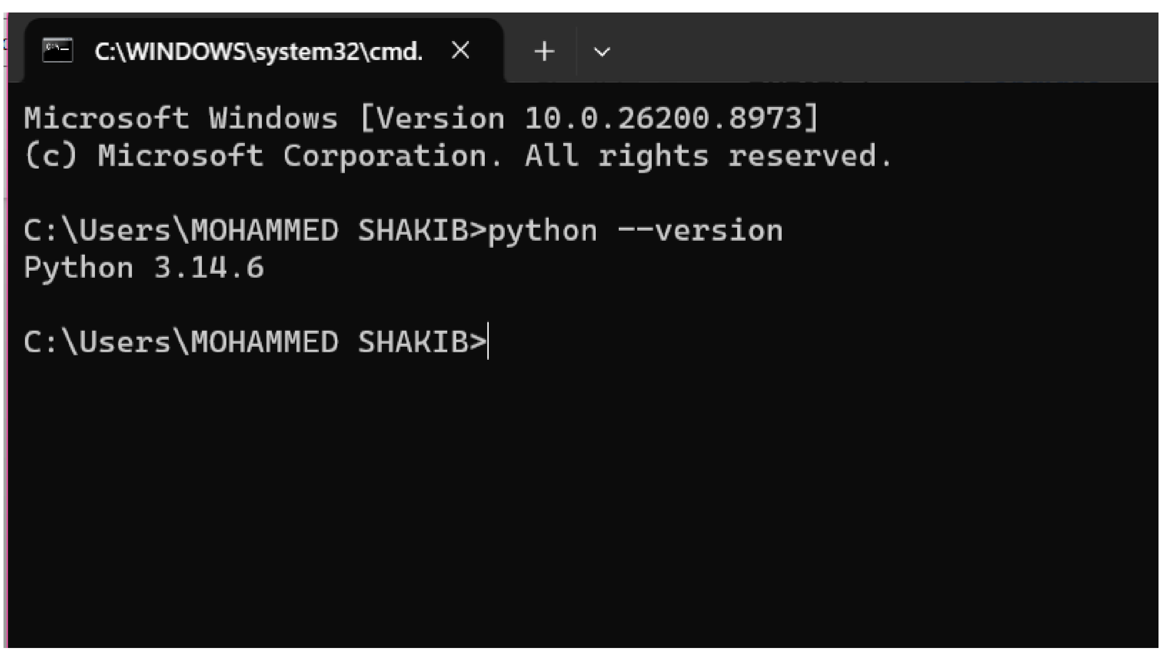
Python and Visual Studio Code were already installed on the system. Jupyter Notebook was configured for interactive Python programming. NumPy and Pandas were installed and successfully tested using Python.

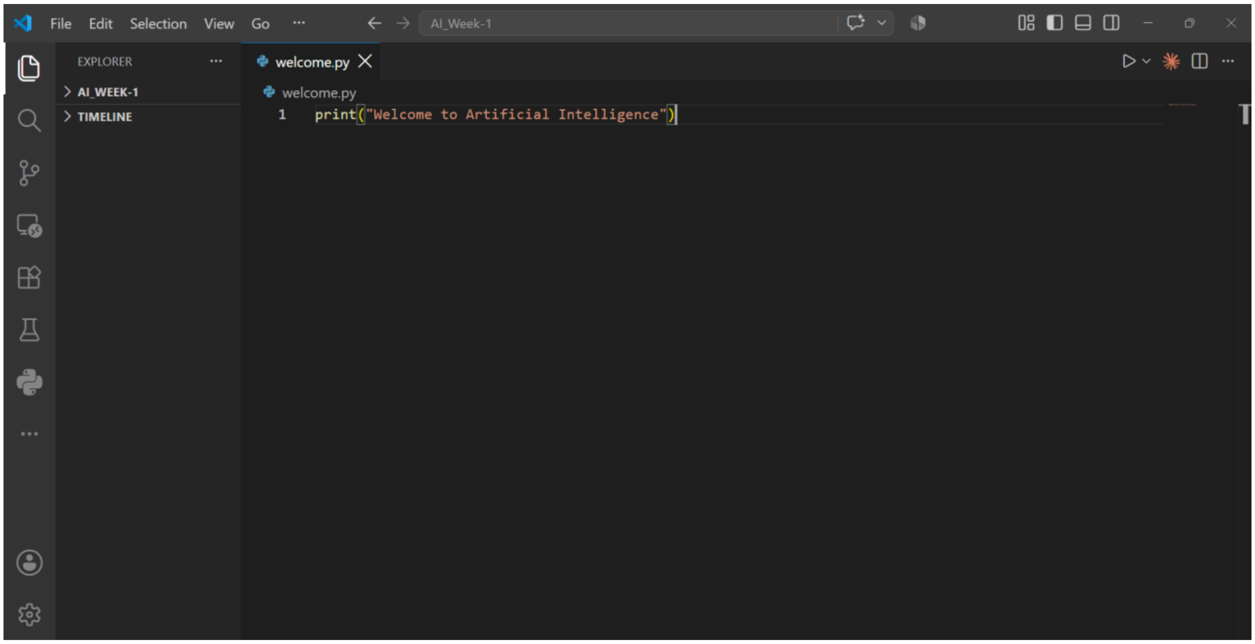
Simple Python Program

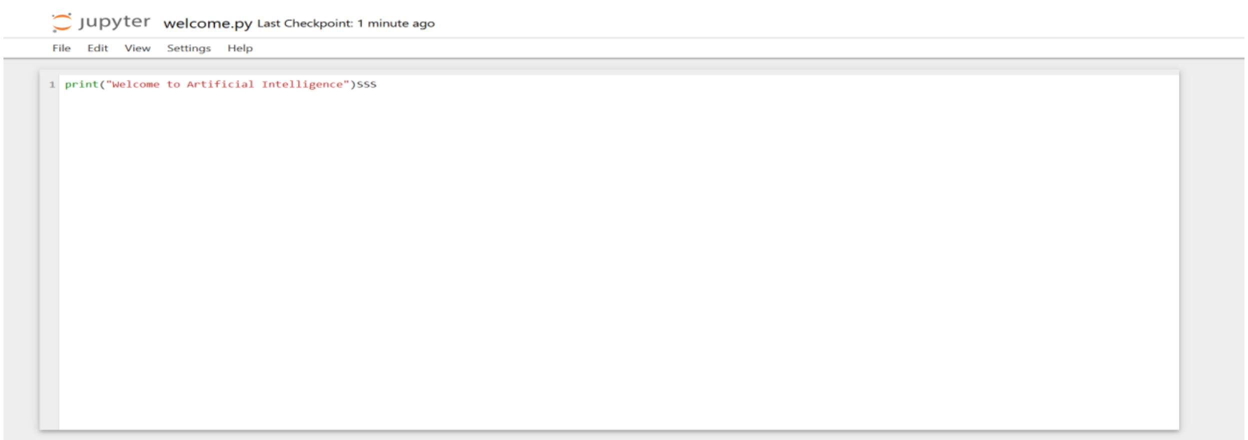
Print ("Welcome to Artificial Intelligence")

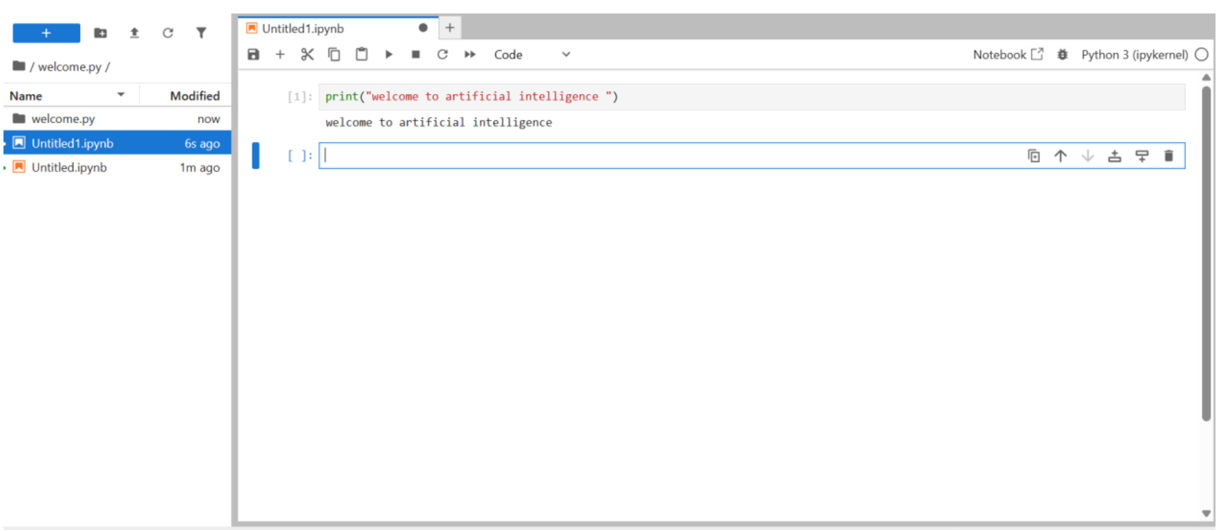
Program Output

Welcome to Artificial Intelligence




File Edit Selection View Go ···
AI_Week-1
EXPLORER
AI_WEEK-1
TIMELINE
welcome.py
welcome.py
1 print("Welcome to Artificial Intelligence")


jupyter welcome.py Last Checkpoint: 1 minute ago
File Edit View Settings Help
1 print("Welcome to Artificial Intelligence")sss


/ welcome.py /
Name
Modified
welcome.py
now
Untitled1.ipynb
6s ago
Untitled.ipynb
1m ago
Untitled1.ipynb
Code
Notebook
Python 3 (ipykernel)
[1]: print("welcome to artificial intelligence ")
welcome to artificial intelligence
[ ]:

Command Prompt

```
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MOHAMMED SHAKIB>python -c "import numpy, pandas; print('NumPy and Pandas are working!')"
NumPy and Pandas are working!

C:\Users\MOHAMMED SHAKIB>
```

Untitled1.ipynb

Code

Notebook Python 3 (ipykernel)

[1]:

```
print("welcome to artificial intelligence ")
```

```
welcome to artificial intelligence
```

[3]:

```
name = "AI Student"
marks = 85
temperature = 32.5
passed = True

print("Name:", name)
print("Marks:", marks)
print("Temperature:", temperature)
print("Passed:", passed)

print("Data type of name:", type(name))
print("Data type of marks:", type(marks))
print("Data type of temperature:", type(temperature))
print("Data type of passed:", type(passed))
```

```
Name: AI Student
Marks: 85
Temperature: 32.5
Passed: True
Data type of name: <class 'str'>
Data type of marks: <class 'int'>
Data type of temperature: <class 'float'>
Data type of passed: <class 'bool'>
```

Task 3: Python for AI

Objective

The objective of this task is to understand basic Python concepts used in Artificial Intelligence programming, including variables, data types, user input, operators, and conditional statements.

Python Concepts Used

1. Variables

Variables are used to store information in a Python program. In this program, name stores the student's name and marks stores the student's marks.

2. Data Types

The program uses a **string** for the student's name and an **integer** for the marks.

3. User Input

The input() function is used to receive the student's name and marks from the user.

4. Operators and Conditional Statement

The >= comparison operator is used to compare the student's marks with 40. An if-else statement then predicts whether the student has passed or failed.

AI-Related Example: Pass/Fail Prediction

Python Program

```
name = input ("Enter your name: ")
marks = int (input ("Enter your marks: "))
print ("\n Student Name:", name)
print ("Marks:", marks)
print ("Data Type of marks:", type(marks))
if marks >= 40:
   result = "Pass"
else:
   result = "Fail"


print ("Prediction:", result)
```

Program Output

For the test case, the student entered the name **Mohammed Shakib** and marks **20**.

Enter your name: mohammed shakib

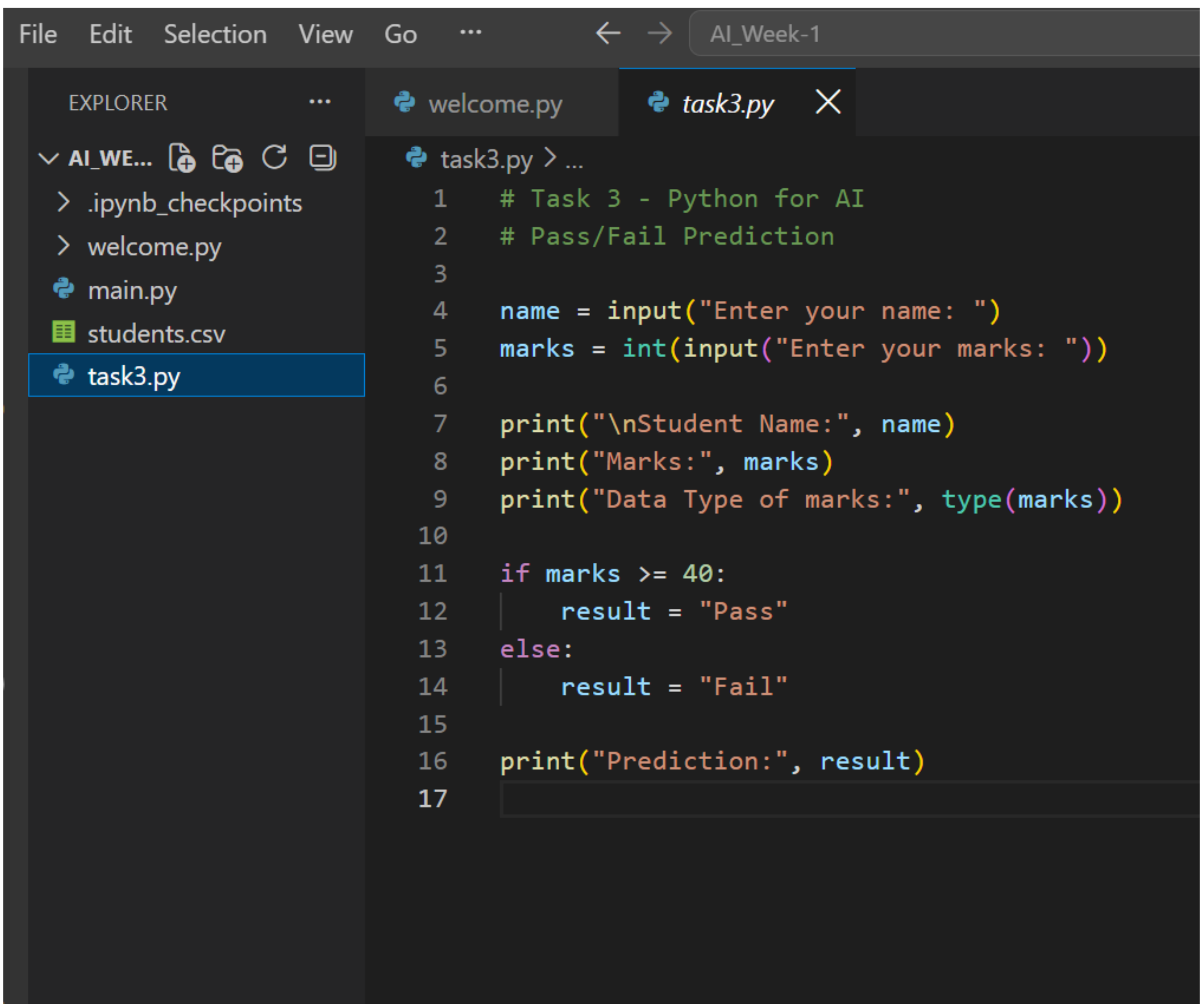
Enter your marks: 100

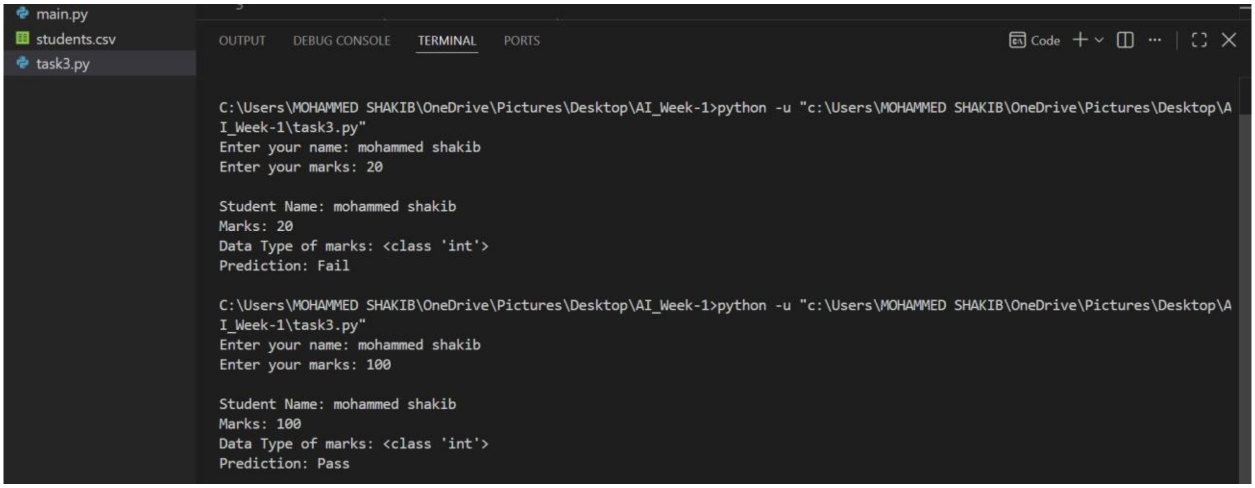
Student Name: mohammed shakib

Marks: 100

Data Type of marks: <class 'int'>

Prediction: pass





Result

The program successfully accepts user input and predicts whether a student has passed or failed based on their marks. This demonstrates how basic Python programming concepts can be used to create a simple rule-based AI-related application.

Task 4: AI Case Study — ChatGPT

Introduction

ChatGPT is an Artificial Intelligence application developed by OpenAI. It is a conversational AI system that can understand natural language and generate responses to user questions and instructions. ChatGPT can be used for education, programming, writing, research, brainstorming, translation, and many other activities.

How AI Is Used

ChatGPT uses Artificial Intelligence techniques such as Natural Language Processing (NLP) and machine learning to understand and generate human-like text. The system is trained on large amounts of data to learn patterns in language. When a user enters a question or instruction, the AI analyzes the input and generates a suitable response based on the context.

Input

The input is the information provided by the user. It can be a question, instruction, sentence, programming code, or other text. For example, a student may enter:

"Explain Artificial Intelligence in simple words."

ChatGPT receives this request and analyzes what the user wants to know.

Processing

During processing, the AI analyzes the words and context of the user's request. Its machine-learning model uses patterns learned during training to determine an appropriate response. It then generates the response based on the user's input and the conversation context.

Output

The output is the response generated by ChatGPT. Depending on the user's request, the output can be an explanation, answer, program code, summary, translation, ideas, or other text.

Benefits of ChatGPT

ChatGPT provides many benefits. It can help students understand difficult topics, assist programmers in solving coding problems, generate ideas, summarize information, and improve writing. It can provide responses quickly and communicate with users using natural language. It can also explain concepts at different levels of difficulty.

Limitations

ChatGPT also has limitations. It can sometimes provide incorrect or outdated information. Therefore, important information should be checked using reliable sources. ChatGPT should be used as an AI assistant and not as a complete replacement for human judgment.

Conclusion

ChatGPT is a practical example of Artificial Intelligence being used in everyday life. It uses machine learning and natural language processing to understand user input and generate useful responses. Its applications in education, programming, communication, and productivity demonstrate the growing importance of AI.

TASK 5: MINI AI PROJECT

Rule-Based AI Chatbot

Introduction

A rule-based chatbot is a simple Artificial Intelligence program that responds to users based on predefined rules. It does not use Machine Learning. Instead, it uses if-elif-else statements to identify the user's message and provide an appropriate response.

Objective

The objective of this project is to create a simple chatbot that can understand basic user messages and respond to greetings, questions about its condition, and goodbye messages.

Technologies Used

- Python
- if-elif -else statements
- input () function
- String comparison

Python Program

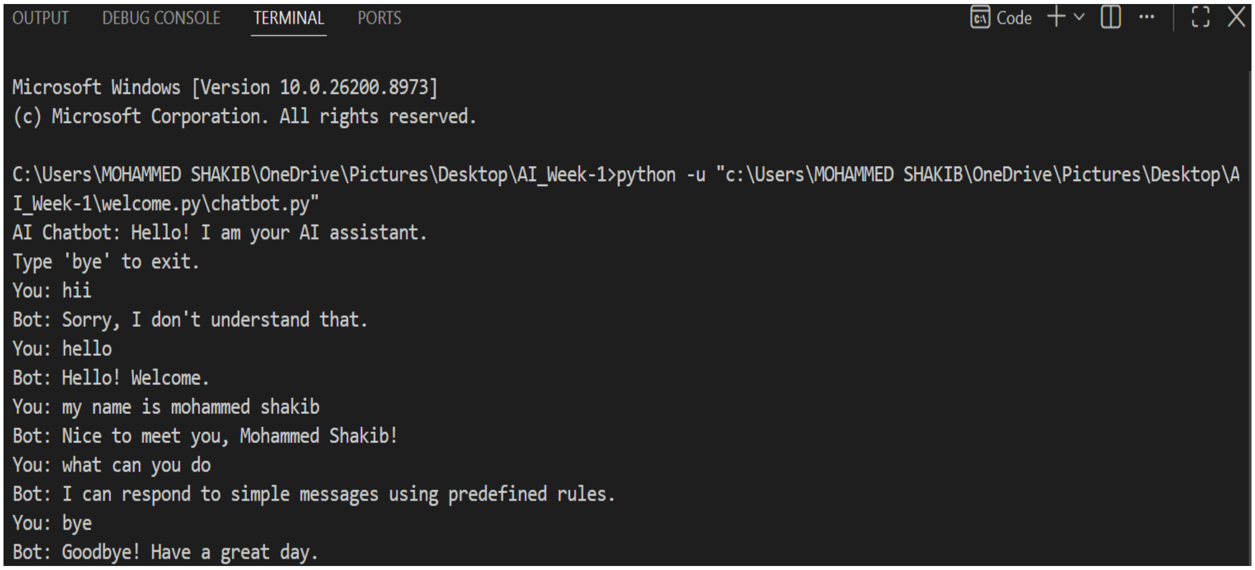
welcome.py chatbot.py

welcome.py > chatbot.py > ...

```
print("AI Chatbot: Hello! I am your AI assistant.")
print("Type 'bye' to exit.")
while True:
    user = input("You: ").lower()
    if user == "hello" or user == "hi":
        print("Bot: Hello! Welcome.")
    elif user == "how are you":
        print("Bot: I'm fine. How are you?")
    elif user.startswith("my name is"):
        name = user.replace("my name is", "").strip()
        print("Bot: Nice to meet you, " + name.title() + "!")
    elif user == "what is your name":
        print("Bot: I am a Rule-Based AI Chatbot.")
    elif user == "what can you do":
        print("Bot: I can respond to simple messages using predefined rules.")
    elif user == "bye":
        print("Bot: Goodbye! Have a great day.")
        break
    else:
        print("Bot: Sorry, I don't understand that.")

```

Program output:



```
OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                  Code

Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MOHAMMED SHAKIB\OneDrive\Pictures\Desktop\AI_Week-1>python -u "c:\Users\MOHAMMED SHAKIB\OneDrive\Pictures\Desktop\AI_Week-1\welcome.py\chatbot.py"
AI Chatbot: Hello! I am your AI assistant.
Type 'bye' to exit.
You: hii
Bot: Sorry, I don't understand that.
You: hello
Bot: Hello! Welcome.
You: my name is mohammed shakib
Bot: Nice to meet you, Mohammed Shakib!
You: what can you do
Bot: I can respond to simple messages using predefined rules.
You: bye
Bot: Goodbye! Have a great day.
```

How It Works

The chatbot takes input from the user using the input () function. The. lower () function converts the input to lowercase so that messages such as "Hello" and "hello" can be treated the same. The if-elif-else statements compare the user's message with predefined rules. If a matching condition is found, the chatbot displays the corresponding response. If no condition matches, the chatbot displays a message saying that it does not understand the input. The while loop keeps the chatbot running until the user enters "bye".

Benefits

- Simple and easy to implement.
- Does not require a Machine Learning model.
- Provides quick responses.
- Useful for understanding the basic concept of chatbots.
- Helps demonstrate rule-based Artificial Intelligence.

Limitations

The chatbot can only respond to messages that have been predefined in its rules. It cannot understand complex questions or learn from conversations. Therefore, it is much simpler than modern AI systems such as ChatGPT.

Result

The Rule-Based AI Chatbot was successfully developed using Python. The project demonstrates how predefined rules and conditional statements can be used to create a basic AI application.

CONCLUSION

This Week 1 assignment provided an introduction to the fundamentals of Artificial Intelligence and Python programming. The different types of AI, including Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Super Intelligence (ASI), were studied along with the differences between AI, Machine Learning, and Deep Learning. A Python-based AI development environment was also configured using Python, Visual Studio Code, Jupyter Notebook, NumPy, and Pandas. Basic Python concepts such as variables, data types, user input, operators, and conditional statements were implemented through a simple pass/fail prediction program.

REFERENCES

1. OpenAI. *ChatGPT*. Official OpenAI website.
2. Python Software Foundation. *Python Documentation*. Python Official Documentation.
3. NumPy Developers. *NumPy Documentation*. NumPy Official Documentation.
4. The pandas development team. *pandas Documentation*. pandas Official Documentation.
5. IBM. *Artificial Intelligence (AI)*. IBM AI resources.
6. Google. *Machine Learning and Artificial Intelligence*. Google AI resources.